

BT401: FINAL TERM- 100% GUESS

PAPER 2023

GENETIC RESOURCES &

CONSERVATION

BY SULMAN ALI

0341-7547044

SUBJECTIVE

Disadvantages of DNA Bank?

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

Briefly explain historical background of Ramsar convention?

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975.

How bottleneck effect cover area?

The bottleneck effect is a sharp lowering of a population's gene pool because of an environmental, or human-caused, change. This can limit adaptability and increase vulnerability to future challenges in that area.

Protection lab in Pakistan for plants:

1= Germplasm exploration lab 2= Seed preservation lab 3= In vitro conservation lab 4= Germplasm evaluation lab 5= Plant introduction and seed health lab 6= Data management lab.

Cartagena protocol on biosafety?

The Cartagena Protocol on Biosafety is an international treaty that aims to ensure the safe handling, transport, and use of genetically modified organisms (GMOs) to protect biodiversity and human health. It promotes information sharing, risk assessment, and regulatory measures among countries to address potential environmental and health risks posed by GMOs.

Features of habitat species management area:

1= Protection of particular species, 2= Protection of habitats, 3= Active management to maintain target species 4= Active management of culturally-defined ecosystems, 5= Zoning, 6= Education and Outreach, 7= Research and Monitoring, 8= Regulations and Enforcement.

Strategies for effective conservation and utilization of AnGR:

1. Formulating the National Livestock and wild-life Breeding Policies. 2. Encouraging the Formation of Breed Associations. 3. Developing Professional Human Resources. 4. Strengthening Research and Development Institutions 5. Developing Infrastructure for marketing

International co-operation and assistance is needed in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources.

Limitation of Germplasm Conservation?

The germplasm conservation through the conventional methods has several limitations such as short-lived seeds, seed dormancy, seed-borne diseases, and high inputs of cost and labor. The techniques of cryopreservation (freezing cells and tissues at -196°C) and using cold storages help us to overcome these problems. Also Germplasm conservation faces challenges of high costs, genetic drift, inadequate representation of diversity, insufficient funding, and the potential inability to address rapidly changing environmental and climatic conditions.

Threats to AnGR:

Despite the importance of animal genetic resources and their diversity, their diversity has been continually decreasing over time.

One of the greatest threats to livestock diversity is pressure from large-scale commercial production systems to maintain only high-output breeds. Changes in climate will have an impact on livestock and food production in many ways.

Some major disease threats that livestock currently face include, rinderpest, foot and mouth disease, and Peste des petits ruminants (PPR), also known as sheep and goat plague.

Historical background of CBD:

The Convention on Biological Diversity, was opened for signature at the Earth Summit in Rio de Janeiro, Brazil, on June 5, 1992 and by the end of July 1993, 165 countries had signed the treaty. The initial sessions were referred to as meetings of the "Ad Hoc Working Group of Experts on Biological Diversity."

How FGR cope with climate change:

Forest genetic resources (FGR) coping with climate change involve selecting and breeding resilient tree varieties, conserving diverse genetic material, promoting assisted migration, and utilizing genomic tools to enhance adaptability and forest health.

Critical Endangered Species?

Angered Species. A critically endangered (CR) species is one which has been categorized by the International Union for Conservation of Nature (IUCN) as facing an extremely high risk of extinction in the wild. Examples: " Eastern Lowland Gorilla " Hawksbill Turtle " Javan Rhino

Role of national park in landscape and seascape?

1= Protecting some of the earth's richness that will not survive outside. 2= Protecting additional ecosystem services. 3= Providing areas where ecosystems can be studied in as pristine an environment as possible. 4= Protecting natural sites that are also of religious and cultural significance.

How animal genetic resources should conserved:

Animal genetic resources are conserved by establishing gene banks, protecting unique breeds in their habitats, practicing sustainable breeding, and collaborating internationally to safeguard genetic diversity, ensuring future food security and adaptation.

Effect of bottleneck effect on alleles frequency:

The bottleneck effect reduces a population's size dramatically, often due to a catastrophic event. This leads to a limited gene pool, causing certain alleles to be lost or become overrepresented, altering the overall allele frequency. This can increase the risk of genetic disorders and reduce genetic diversity, potentially impacting the population's ability to adapt to environmental changes.

An example of the bottleneck effect on allele frequency is the cheetah population. After a severe reduction in numbers, cheetahs experienced a genetic bottleneck, resulting in low genetic diversity and a higher frequency of certain alleles. This has made them more susceptible to diseases and environmental changes, highlighting the potential negative consequences of reduced genetic variation due to a bottleneck event.

What Is Botanical Garden?

It is a garden dedicated to the collection, cultivation and display of a wide range of Plants labeled with their botanical names. It may contain specialist plant collections such As tropical plants, or other species of plants.

Which type of material is used in DNA bank?

DNA banks typically use special tubes or plates made of materials like plastic or glass to store DNA samples. These materials keep the DNA safe and stable over time. Most DNA provided by DNA banks is used for studies to attempt to develop more productive or more environmentally friendly agricultural species. Some DNA banks also store the DNA of rare or endangered species to ensure their survival. The DNA bank can be used to compare and analyse DNA samples.

Forest genetic resources:

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

Working of Storage in DNA banks:

DNA banks store DNA samples in controlled environments, using cold temperatures and special preservatives. This prevents degradation and maintains the genetic material's integrity for future research or analysis.

What are the factors or conditions for a national park?

National parks are designated areas of natural significance that must meet specific criteria. These include unique landscapes, ecosystems, or wildlife, along with opportunities for public enjoyment and education. Preservation of natural beauty and cultural heritage, ecological balance, and sustainable recreation are vital. Adequate size and isolation to maintain biodiversity, and legal protection to prevent exploitation are necessary. Scientific research and conservation efforts contribute to their value as well.

Define microbial genetic resources programs working in Pakistan

To establish a long term microbial preservation facility as “National Culture Collection of Pakistan (NCCP)” for collection and preservation of microbial genetic resources from Pakistani ecology for basic and applied microbiology and biotechnology research following the Best Practice Guidelines of OECD (2001).

To distribute the economically important strains to scientific community, institutions and industry on commercial basis for research or utilization in industrial processes as a regular activity of the NCCP by following national and international laws and regulations.

Category V: Protected Landscape-Seascape:

A protected area where the interaction of people and nature over time has produced an area of distinct character with significant ecological, biological, cultural and scenic value. It's endorsed by the International Union for Conservation of Nature (IUCN) in collaboration with bodies like the World Commission on Protected Areas and national governments. The key objective is preserving and nurturing these vital landscapes/seascapes alongside their linked nature conservation. This category encompasses various goals, from promoting sustainable interactions of nature and culture, supporting conservation, enabling recreation, and economic activities to fostering community engagement, conserving biodiversity, and setting examples of

sustainability. These areas boast high scenic value, rich habitats, and a harmonious blend of human and natural influences.

What is the role of phenotypic plasticity to the environmental changes?

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions. This adaptation allows organisms to better suit their changing surroundings without requiring genetic changes. Phenotypic plasticity plays a crucial role in helping species cope with shifting conditions, enhancing their survival and reproduction. It allows for short-term adjustments in physiology, behavior, and morphology, aiding species' resilience and success in dynamic and unpredictable environments.

The Sardine Run:

The Sardine Run is a significant migration off South Africa's eastern coast. In July, after spawning in the Cape's cold waters, millions of sardines move north to warmer waters along the KwaZulu-Natal coast, creating a vital ecological event and attracting predators in a spectacular display of marine life.

Hingol National Park:

The conservation status: Vulnerable

Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, Balochistan • It is the largest of National Parks of Pakistan • The area was for the first time declared reserved in 1988. • Hingol is known to support threatened invertebrates in addition to a variety of bird species. The park is an excellent habitat to wild animals including over 3000 ibexes, and 1500 Urials and more than 1200 Chinkara • A number of resident and migratory birds are supported by this park.

Why cryoprotectants are important for cryopreservation:

Cryoprotectants are crucial for cryopreservation as they prevent cellular damage during freezing by reducing ice crystal formation, osmotic stress, and toxic effects. They maintain cell integrity, enabling successful preservation and eventual revival of biological materials.

Three Botanical Gardens?

1= Cacti and succulent plants. 2= Herb gardens 3= Greenhouses, shade houses. 4= Tropical plants. 5= Medicinal Plants. 6= Aromatic or textile plants 7= Other exotic plants.

In vivo Gene Bank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources.

What Is the Extreme Weather Effect on Invertebrate's Genetic Resources?

Extreme weather can impact invertebrate genetic resources through habitat destruction, altered breeding patterns, and genetic mutations. Such events can lead to genetic diversity loss, affecting ecosystem stability and overall biodiversity.

why plants are threatened in Pakistan:

Plants in Pakistan face threats from deforestation, habitat degradation, urbanization, and climate change. Unsustainable land use practices and inadequate conservation efforts contribute to their vulnerability and decline.

Define sessional migration:

Seasonal migration refers to the phenomenon where certain animals, such as birds or whales, regularly move from one region or habitat to another in response to seasonal changes in resources or environmental conditions.

Obstacles in Cryopreservation?

Obstacles Upto 60% human body is composed of water. What's the issue then? • Actually the freezing point of water is 0 degree centigrade while the cryoscopy temperature can be as low as - 90 degree centigrade. • Very expensive Technique • Ice formation can result in the needle shaped crystals resulting in the damage to cell membrane. • Unequal distribution or over distribution of cryoprotectants. • Moreover, thermal gradients can induce mechanical stress due to uneven expansion or contraction in the biomaterial. • The cooling rate required for optimal survival varies by several orders of magnitude between different cell types. • Mass transfer limitations.

Feature of Pakistan terrestrial water and maritime act?

It includes provisions for preservation, development and protection of marine environment.

This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ) •

This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

Difference b/w law and legislation?

Law refers to a system of rules and principles governing a society, while legislation specifically refers to the process of creating and enacting laws by a legislative body, like a parliament or congress.

Primary objective of Category VI?

To protect natural ecosystems. Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

Cartagena Protocol areas

The Protocol covers: • Transboundary movement, transit, handling and use of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

It does not cover: • Products derived from LMOs (e.g. paper from GM trees) • LMOs, which are pharmaceuticals for humans that are addressed by other relevant international agreements or organizations.

Strict Nature Reserve:

Category Ia are strictly protected areas set aside to protect biodiversity and also possibly geological/geomorphic features, where human visitation, use and impacts are strictly controlled and limited to ensure protection of the conservation values. Such protected areas can serve as indispensable reference areas for scientific research and monitoring.

Close Inbreeding:

Close inbreeding refers to the breeding of individuals that are closely related, such as siblings or parent-offspring pairs. This type of inbreeding increases the risk of genetic disorders and reduces genetic diversity within a population.

Line Breeding:

Mating animals that are more distantly related which can be traced back to one common ancestor. e.g. Cousins Grandparents to grand offspring, Half-brother to half-sister. Line breeding increases genetic purity amongst the animals of progeny generations.

What Is Cold Storage. Discuss?

Cold Storage - it involves storage in low and nonfreezing temperature.

Slow Freezing - Slow Thawing:

Organs or embryos are treated with a protective solution, slowly cooled to -1°C at a rate of $0.5\text{--}2^{\circ}\text{C}$ per minute, followed by seeding and a brief equilibration period. Afterward, they're further cooled to -60°C before being stored in liquid nitrogen. Thawing must be slow to prevent damage.

Rapid Cooling and Rapid Thawing: Cooling stops at -30 to -40°C , and embryos are rapidly cooled in liquid nitrogen to -196°C . Thawing is quick ($200\text{--}500^{\circ}\text{C}$ per minute) to avoid forming damaging crystals.

Vitrification: Vitrification turns tissue water into a glass-like state instead of crystals by increasing viscosity. This prevents harmful ice crystal formation during freezing.

Ultra-Rapid Freezing: Embryos are quickly treated with high-concentration cryoprotectant, then immersed in liquid nitrogen. Thawing is done with a warm water bath (about 500°C per minute).

Application for permit to import plant material:

- 1) Before any plant or plant material is imported, an application for permit shall be submitted to the Director or to the Entomologist (Quarantine).
- 2) All such applications shall be signed by the person who intends to import the plant or plant material or his duly authorized agent and shall specify:
 - (a) the kind and quantity of plant or plant material; (b) the country and locality of origin; (c) destination; (d) the name and address of the consignor and the consignee; (e) means of transport; (f) the prescribed port or point of entry; (g) the prescribed for which the plant or plant material is proposed to be imported e.g., consumption propagation or processing.

Name the Physical growth limitation:

1= Low temperature 2= Low light/restricted photo period 3= Minimal containment 4= Minimal O₂ 5= Osmotic (water) stress.

What are national parks also define their purpose:

National parks are protected areas of natural significance, preserving diverse ecosystems, wildlife, and landscapes for present and future generations. Their purpose is to conserve biodiversity, offer recreational opportunities, and promote education and research while safeguarding the environment from human impacts.

Conserve plant biodiversity:

"Conserve plant biodiversity" refers to preserving the variety and abundance of plant species in their natural habitats, ensuring their survival, genetic diversity, and ecological roles for the health of ecosystems and humanity.

Possible outcomes of introduction of exotic species in FGR:

The introduction of exotic species in forest genetic resources (FGR) can result in adverse consequences, such as displacement of indigenous species, disruption of ecosystem equilibrium, genetic contamination, reduced adaptability, and ecosystem instability, posing risks to local biodiversity and long-term forest sustainability.

Who Is Responsible for A Botanical Gardens?

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

What Are Three Pillars of Ramsar Convention?

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In 18 nations meeting it was adopted in the Iranian city of Ramsar on 2nd February 1971. Came into force in 21st December 1975.

The “three pillars” of the convention.

The Contracting Parties (160) commit to: 1= Work towards the wise use of all their wetlands. 2= Designate suitable wetlands for the list of Wetlands of International Importance. 3= Cooperate internationally on trans boundary wetlands, shared wetland systems and shared species.

Difference Between Genetic Drift and Gene Flow?

Genetic drift involves random changes in allele frequencies within small populations due to chance events, potentially leading to significant genetic variations over time. Gene flow, on the other hand, refers to the exchange of genetic material between different populations through migration or interbreeding. It promotes genetic diversity, counteracting the effects of drift by introducing new alleles. While drift is driven by stochastic events, gene flow is facilitated by movement and reproduction, influencing the genetic makeup and adaptability of populations.

Objectives of Cartagena Protocol on Biosafety:

The Cartagena Protocol on Biosafety aims to ensure the safe handling, transport, and use of genetically modified organisms (GMOs) to protect biodiversity and human health. Its objectives include minimizing potential risks associated with GMOs, promoting public awareness, facilitating international cooperation, and providing a framework for transparent and informed decision-making regarding GMOs.

Application of Cryopreservation In Animal Husbandry.

Cryopreservation in animal husbandry conserves genetic diversity, aids endangered species recovery, and supports assisted reproduction by freezing and storing gametes, embryos, or tissue, ensuring future breeding and research opportunities.

Cryopreservation in plants?

- Selection of material
- Addition of cryo-protectant
- Freezing
- Storage in liquid nitrogen
- Thawing
- Washing and re-culturing
- Measurement of viability
- Regeneration of plants

Physical factor in medium conservation:

Temperature is a critical physical factor in medium conservation, influencing chemical reactions, microbial growth, and degradation rates, determining the stability and preservation of the medium's components.

Note on types of genetics resources:

Genetic resources include plant, animal, and microbial diversity. Plant genetic resources encompass crops, wild relatives, and landraces. Animal genetic resources involve livestock breeds, while microbial resources encompass microorganisms for various applications.

The Services Provided By PEPA?

1= Protection 2= Conservation, 3= Rehabilitation 4= Improvement of the environment; 5= PEPA provide framework for prevention and control of pollution 6= Helps in protection of sustainable development.

Criteria of National Park?

Distinguishing Features ♣ The area should contain representative examples of major natural regions, and biological and environmental features or scenery. ♣ It should be of sufficient size to maintain ecological processes. ♣ The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

State Quarantine Principle?

The quarantine principle involves restricting the movement of potentially harmful organisms, materials, or goods to prevent the introduction and spread of pests, diseases, or contaminants across borders, safeguarding ecosystems and industries.

How many ways bacteria have transfer their DNA horizontally?

Horizontal gene transfer is basically the transfer of genes between organisms via methods other than asexual or sexual reproduction. Genes and the characteristics code for are passed down from parent to progeny. There are three ways for bacteria to transfer their DNA horizontally.

Conjugation: The transfer of DNA directly from one cell to another through cell-cell contact often involving plasmids.

Transformation: Bacteria are capable of taking up DNA directly from their environment and incorporating it into their genomes known as natural transformation.

Transduction: Transduction is the transfer of DNA from one cell to another by a virus.

What is the Game reserve:

A game reserve is a designated area set aside for the conservation and protection of wildlife, where hunting and other human activities are controlled or prohibited. It aims to preserve biodiversity, provide habitats for animals, and offer opportunities for ecotourism and education.

Biodiversity of Habitat Species Management Area:

Definition: IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as “protected areas aiming to protect particular species or habitats and management reflect this priority.

Role of microorganisms in agriculture:

Microorganisms play a vital role in agriculture by enhancing soil fertility through nutrient cycling, promoting plant growth, fixing nitrogen, decomposing organic matter, controlling pests, and supporting sustainable farming practices.

Write the Variations In Variation Gene?

There are three primary sources of genetic variation, which we will learn more about:

♥ Mutations are changes in the DNA. ♥ Gene flow is any movement of genes from one population to another and is an important source of genetic variation ♥ Sex can introduce new gene combinations into a population.